Chapter 19

Redox Reactions & Electrochemistry

Sections covered: 19.1 Redox Reactions · 19.2 Electrochemical Reactions. All content sourced from your Chapter 19 slide deck. Symbols: → (yields/produces) · ⇌ (equilibrium) · • (bullet)

CONTENTS

01 Complete Vocabulary — All 16 Terms

oxidation-reduction reaction (redox reaction)
A class of chemical reactions that involves the transfer of electrons from one substance to another. Oxidation and reduction always occur simultaneously — you cannot have one without the other. Only reactions in which oxidation numbers change are considered redox reactions.

oxidation
The process by which an atom or ion LOSES one or more electrons. When oxidation occurs, the oxidation number of that atom becomes MORE POSITIVE. Example: $Mg \rightarrow Mg^{2+} + 2e^-$ (magnesium loses 2 electrons, goes from 0 to +2).

reduction
The process by which an atom or ion GAINS one or more electrons. When reduction occurs, the oxidation number of that atom becomes MORE NEGATIVE (smaller). Example: $O_2 + 4e^- \rightarrow 2O^{2-}$ (oxygen gains electrons, goes from 0 to −2).

reducing agent
A substance that causes the REDUCTION of another substance in a redox reaction. The reducing agent donates its electrons to the other substance. Important: reducing agents are always OXIDIZED themselves in redox reactions — they give away electrons, so their own oxidation number increases. Example in $2Na + 2H_2O \rightarrow 2NaOH + H_2$: sodium is the reducing agent (it gets oxidized).

oxidizing agent
A substance that causes the OXIDATION of another substance in a redox reaction. The oxidizing agent accepts electrons from the other substance. Important: oxidizing agents are always REDUCED themselves in redox reactions — they receive electrons, so their oxidation number decreases. Example in $2Na + 2H_2O \rightarrow 2NaOH + H_2$: hydrogen in water is the oxidizing agent (it gets reduced).

electrochemistry
The field of chemistry that studies how electricity and redox reactions are related. It covers two related processes: (1) spontaneous redox reactions that produce electrical current, and (2) electrical current that drives nonspontaneous redox reactions.

electrolyte
Any substance that, when dissolved in water, allows the resulting solution to conduct electricity. A substance's ability to act as an electrolyte depends on its ability to form ions. Strong electrolytes (most salts, strong acids, strong bases) produce many ions and conduct well. Weak electrolytes produce fewer ions and conduct poorly.

electrochemical cell
A device that uses either a spontaneous chemical reaction to produce electricity, or uses electricity to drive a nonspontaneous chemical reaction. There are two types: voltaic cells (produce electricity) and electrolytic cells (consume electricity).

voltaic cell
An electrochemical cell that uses a SPONTANEOUS chemical reaction to produce electrical energy. Energy flows from chemical → electrical. The reaction occurs on its own without any external power source. Examples: batteries (AA, car battery, lithium-ion), fuel cells.

electrolysis
The process of forcing an otherwise nonspontaneous redox reaction to occur with the aid of an electrical current in an electrochemical cell. The electrical energy drives a reaction that would not happen on its own. Used in electroplating, aluminum smelting, and purifying active metals.

cathode
The electrode where REDUCTION occurs in any electrochemical cell. Memory tip: Red Cat (REDuction at CATHode). Charge: NEGATIVELY charged in electrolytic cells; POSITIVELY charged in voltaic cells. Cations (positive ions) migrate toward the cathode.

anode
The electrode where OXIDATION occurs in any electrochemical cell. Memory tip: An Ox (ANOde = OXidation). Charge: POSITIVELY charged in electrolytic cells; NEGATIVELY charged in voltaic cells. Anions (negative ions) migrate toward the anode.

electrodeposition / electroplating
Electrodeposition is any process that deposits one substance onto another by an electrochemical reaction. Electroplating is a specific type of electrolysis in which the metal object to be plated is the CATHODE (it receives metal ions → reduction deposits metal on it), and the metal that is to cover the object is the ANODE (it dissolves → oxidation).

half-cell
One of the two compartments in a voltaic cell. One half-cell contains the anode where oxidation takes place; the other contains the cathode where reduction takes place. The two half-cells are connected by a salt bridge that allows ion flow.

half-reaction
The oxidation or reduction portion of a redox reaction written as a separate equation, showing the electrons explicitly. Two half-reactions (one oxidation + one reduction) are combined to form the complete balanced redox reaction. Example: $Zn \rightarrow Zn^{2+} + 2e^-$ (oxidation half-reaction).

salt bridge
A tube of electrolytic gel (or a porous membrane) that connects the two half-cells of a voltaic cell. Its purposes: (1) allows the flow of ions between the two electrolytic solutions, (2) prevents the two solutions from mixing, (3) maintains electrical neutrality in each half-cell so the reaction can continue. Without a salt bridge, ion buildup would stop the reaction.

fuel cell
A type of voltaic cell that generates electricity from a continuous supply of fuel (commonly hydrogen) and an oxidant (oxygen from air). Unlike regular batteries, fuel cells do not become depleted — as long as fuel is supplied, electricity is generated. The only byproduct of a hydrogen fuel cell is water.

02 Section 19.1: Oxidation & Reduction — Core Concepts

Fundamental principle: Oxidation and reduction always occur simultaneously in a redox reaction. The electrons that leave one atom (oxidation) must join another atom (reduction). You cannot have oxidation without reduction, and vice versa.

> How to identify a redox reaction: Assign oxidation numbers to all atoms in the equation. If ANY atom changes its oxidation number from reactants to products, it is a redox reaction. If all oxidation numbers stay the same, it is NOT a redox reaction.

Memory trick — OIL RIG:

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OIL RIG:

Oxidation Is Loss (of electrons)

Reduction Is Gain (of electrons)
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What happens to oxidation numbers:

Process	Electrons	Oxidation Number	Example
Oxidation	LOST (given away)	Becomes MORE POSITIVE	Mg: 0 → +2
Reduction	GAINED (received)	Becomes MORE NEGATIVE	O: 0 → −2

Example — $2Mg + O_2 \rightarrow 2MgO$

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2Mg + O₂ → 2MgO
```

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0 0 +2 −2

Mg: 0 → +2 (OXIDIZED — lost 2 electrons each)

O: 0 → −2 (REDUCED — gained 2 electrons each)
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03 Oxidizing & Reducing Agents

> Critical distinction: The NAME of an agent describes what it does TO OTHERS, NOT what happens to itself. A reducing agent causes reduction in others, but is itself oxidized. An oxidizing agent causes oxidation in others, but is itself reduced.

Agent	What it does TO OTHERS	What happens TO IT	Electron direction
Reducing agent	Causes REDUCTION of another substance	Gets OXIDIZED (loses e^-)	Donates electrons
Oxidizing agent	Causes OXIDATION of another substance	Gets REDUCED (gains e^-)	Accepts electrons

Example — $2Na + 2H_2O \rightarrow 2NaOH + H_2$

```
2Na + 2H₂O → 2NaOH + H₂

0 +1 +1 0

Na: 0 → +1 → Na is OXIDIZED → Na is the REDUCING AGENT

H: +1 → 0 → H is REDUCED → H₂O is the OXIDIZING AGENT
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Example — $2KBr + Cl_2 \rightarrow 2KCl + Br_2$ (from section review)

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2KBr + Cl₂ → 2KCl + Br₂

−1 0 −1 0

Br: −1 → 0 → Br is OXIDIZED → KBr is the REDUCING AGENT

Cl: 0 → −1 → Cl is REDUCED → Cl₂ is the OXIDIZING AGENT
```

04 Assigning Oxidation Numbers — Rules & Examples

Rules for assigning oxidation numbers (in order of priority):

1. Free elements: Any element in its pure form (like O_2, Fe, Na, Cl_2) has an oxidation number of 0.
2. Monatomic ions: The oxidation number equals the ion’s charge. $Na^+ = +1$, $Cl^- = -1$, $Fe^{3+} = +3$.

3. Oxygen: Usually −2, except in peroxides (e.g., H_2O_2) where it is −1, and in compounds with fluorine where it is positive.

4. Hydrogen: Usually +1 when bonded to nonmetals; −1 when bonded to metals (metal hydrides).

5. Fluorine: Always −1 (most electronegative element).

6. Neutral compounds: The sum of all oxidation numbers must equal 0.

7. Polyatomic ions: The sum of all oxidation numbers must equal the charge of the ion.

Examples from section review:

Compound / Ion	Oxidation Numbers
F_2	F = 0 (free element)
$Cu(NO_3)_2$	Cu = +2, NO_3^- = −1 each
Cr^-	Cr = −1 (monatomic ion)
$Fe(OH)_2$	Fe = +2, OH = −1 each
H_2S	H = +1, S = −2
CH_2O_2	C = +2, H = +1, O = −2

05 Balancing Redox Reactions — The 5-Step Method

> What must be balanced in a redox equation: (1) Number of each kind of atom on each side. (2) Overall charge on each side. (3) The number of electrons MUST be conserved — electrons lost by the oxidized substance = electrons gained by the reduced substance.

Step 1: Assign oxidation numbers to all atoms

Identify all atoms in the equation and assign oxidation numbers. Focus on finding the atoms whose oxidation numbers change — these are the ones involved in the redox process.

Step 2: Identify which atoms are oxidized and which are reduced

Compare the oxidation numbers of each atom in the reactants vs. products. An increase in oxidation number = oxidation. A decrease = reduction.

Step 3: Draw lines; calculate the net change in oxidation number

Connect the oxidized atoms on the reactant side to the oxidized atoms on the product side, and do the same for reduced atoms. Write the magnitude of the change along each line. This number represents the electrons transferred.

Step 4: Cross-multiply to balance electron transfer

Multiply each reactant-product pair by coefficients such that the total electrons lost (from oxidation) equals the total electrons gained (from reduction). Use the other change as the coefficient: if oxidation change = +3 and reduction change = −4, multiply the oxidation pair by 4 and the reduction pair by 3 (so both = 12 electrons).

Step 5: Finish balancing by inspection

With the redox-involved atoms balanced, use conventional inspection (trial and error) to balance the remaining atoms (usually H and O). Check that atoms AND charges balance on both sides.

06 Worked Examples 19-1 through 19-4

Example 19-1: Is it a redox reaction?

(a) Reaction: $2KBr(aq) + Cl_2(g) \rightarrow 2KCl(aq) + Br_2(l)$

```
K: +1 → +1 (no change)

Br: −1 → 0 (CHANGED — oxidation)

Cl: 0 → −1 (CHANGED — reduction)

→ YES, this is a redox reaction.
```

(b) Reaction: $2NaOH(aq) + CuCl_2(aq) \rightarrow Cu(OH)_2(s) + 2NaCl(aq)$

```
Na: +1 → +1, OH: −1 → −1, Cu: +2 → +2, Cl: −1 → −1

→ NO changes in oxidation numbers → NOT a redox reaction (it is a double-displacement).
```

Example 19-2: Identifying agents

Reaction: $2Na + 2H_2O \rightarrow 2NaOH + H_2$

```
Na: 0 → +1 (oxidized → Na is the REDUCING AGENT)

H: +1 → 0 (reduced → H₂O is the OXIDIZING AGENT)

Sodium was oxidized (lost electrons).

Hydrogen in water was reduced (gained electrons).
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Example 19-3: More complex identification

Reaction: $2S_2O_3^{2-} + OCl^- + 2H^+ \rightarrow Cl^- + S_4O_6^{2-} + H_2O$

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S in S₂O₃²⁻: −2 → product S₄O₆²⁻: oxidation number increases → OXIDIZED

Cl in OCl⁻: +1 → Cl⁻: −1 → REDUCED

Thiosulfate ion (S₂O₃²⁻) is the REDUCING AGENT.

Hypochlorite ion (OCl⁻) is the OXIDIZING AGENT.
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Example 19-4: Full 5-step balancing

Starting equation: $HCl + HNO_3 \rightarrow HOCl + NO + H_2O$

Step 1 — Assign oxidation numbers (changed atoms only):

Cl in HCl: −1 Cl in HOCl: +1

N in HNO_3: +5 N in NO: +2

Step 2 — Identify:

Cl: −1 → +1 (OXIDIZED, change = +2)

N: +5 → +2 (REDUCED, change = −3)

Step 3 — Net changes:

Oxidation change: +2 | Reduction change: −3

Step 4 — Cross-multiply to balance electrons:

+2 × 3 = +6 and −3 × 2 = −6 ✓ (6 electrons each)

$3HCl + 2HNO_3 \rightarrow 3HOCl + 2NO + H_2O$

Step 5 — Inspect:

H: left = 3+2 = 5, right = 3+0+2 = 5 ✓

O: left = 6, right = 3+1+1 = 5 ... adjust → add H_2O:

FINAL: $3HCl + 2HNO_3 \rightarrow 3HOCl + 2NO + H_2O$ ✓

Section Review 19.1 Balancing answers (problem 8):

a. $SnCl_4 + Fe \rightarrow SnCl_2 + FeCl_3$

Balanced: $3SnCl_4 + 2Fe \rightarrow 3SnCl_2 + 2FeCl_3$

b. $SO_2 + Br_2 + H_2O \rightarrow HBr + H_2SO_4$

Balanced: $SO_2 + Br_2 + H_2O \rightarrow 2HBr + H_2SO_4$

c. $CO + I_2O_5 \rightarrow I_2 + CO_2$

Balanced: $5CO + I_2O_5 \rightarrow I_2 + 5CO_2$

07 Section 19.2: Electrochemistry — Core Concepts

Electrochemistry studies two related processes:

1. Spontaneous redox reactions that produce electrical current (voltaic cells).
2. Electrical current that drives nonspontaneous redox reactions (electrolytic cells).

Electrolytes: Substances that allow solutions to conduct electricity by forming ions. Strong electrolytes (strong acids, strong bases, most salts) produce many ions and conduct electricity very well. Weak electrolytes produce fewer ions.

Why can active metals like aluminum and sodium only be purified by electrolysis? Active metals are held in much stronger chemical bonds than less active metals. Additionally, there are very few other elements that are chemically active enough to replace these metals in their compounds. Because of this, standard chemical reduction cannot work — only electrolysis (using electrical energy to force the reaction) can purify them. This is why aluminum smelting and the electrolysis of brine are important industrial processes.

08 Voltaic vs Electrolytic Cells — Full Comparison

Property	Voltaic Cell	Electrolytic Cell
Reaction type	SPONTANEOUS	NONSPONTANEOUS
Energy direction	Chemical → Electrical	Electrical → Chemical
What it does	Produces electrical energy from reaction	Uses electrical energy to force reaction
Anode charge	NEGATIVE (−)	POSITIVE (+)
Cathode charge	POSITIVE (+)	NEGATIVE (−)
Anode process	Oxidation (always)	Oxidation (always)
Cathode process	Reduction (always)	Reduction (always)
Requires external power?	No — generates its own	Yes — must supply electricity
Examples	AA battery, car battery, lithium-ion, fuel cell	Electroplating, aluminum smelting, electrolysis of brine

> KEY TRICK — Charge flip between cell types: Oxidation ALWAYS occurs at the anode and reduction ALWAYS occurs at the cathode in BOTH cell types. BUT the charge on each electrode is OPPOSITE between the two types. In a voltaic cell: anode is (−) and cathode is (+). In an electrolytic cell: anode is (+) and cathode is (−). This is the #1 trick question.

Memory tricks:

```
Red Cat · An Ox
REDuction at CATHode
ANOde = OXidation

OIL RIG (for redox generally):
Oxidation Is Loss · Reduction Is Gain
```

Electrolysis (electrolytic cells):

Electrolysis is the process of forcing an otherwise nonspontaneous redox reaction to occur using an electrical current. The electrical energy provides the energy needed to drive the reaction in the non-spontaneous direction. Applications:

• Electroplating: Coating objects with a thin layer of metal (object = cathode, plating metal = anode).

• Aluminum smelting: Purifying aluminum from bauxite ore using electrolysis.

• Electrolysis of brine: Producing chlorine gas and sodium hydroxide from salt water.

09 Electrodes: Cathode & Anode In Depth

Electrode	Process	Voltaic charge	Electrolytic charge	Ion attracted	Electron flow
Anode	Oxidation (loses e^-)	Negative (−)	Positive (+)	Anions (− ions)	Electrons flow OUT of anode
Cathode	Reduction (gains e^-)	Positive (+)	Negative (−)	Cations (+ ions)	Electrons flow INTO cathode

In a voltaic cell (like the Zn-Cu battery from the lab):

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Anode (Zn, negative): Zn → Zn²⁺ + 2e⁻ (Zn oxidizes, dissolves)
Cathode (Cu, positive): Cu²⁺ + 2e⁻ → Cu (Cu²⁺ is reduced, deposits)
Electrons flow from Zn anode → through wire → to Cu cathode.
Salt bridge: anions flow toward Zn side, cations flow toward Cu side.
Overall: Zn + Cu²⁺ → Zn²⁺ + Cu Voltage ≈ 1.1 V
```

In an electrolytic cell (electroplating):

The object to be plated is made the cathode. Metal ions from the solution are attracted to the cathode and reduced there, depositing a layer of metal. The anode is made of the plating metal itself, which oxidizes and dissolves into the solution, replenishing the metal ions.

10 Half-Reactions — Writing & Balancing

Every voltaic cell has two half-cells. The reaction in each half-cell is called a half-reaction. A half-reaction shows either the oxidation or the reduction step separately, with electrons written explicitly.

How to write and balance half-reactions:

1. Write the oxidation half-reaction: show the species losing electrons and put the electrons on the PRODUCT side.
2. Write the reduction half-reaction: show the species gaining electrons and put the electrons on the REACTANT side.
3. Find the LCM of the electrons in each half-reaction.
4. Multiply each half-reaction by the appropriate factor so both have the same number of electrons.
5. Add the two half-reactions together, canceling the electrons (they must appear in equal numbers on both sides).

Example 19-5: Ca oxidation + Fe^{3+} reduction

Half-reactions as written:

Oxidation: $Ca \rightarrow Ca^{2+} + 2e^-$

Reduction: $Fe^{3+} + 3e^- \rightarrow Fe$

LCM of 2 and 3 = 6 electrons. Multiply:

×3: $3Ca \rightarrow 3Ca^{2+} + 6e^-$

×2: $2Fe^{3+} + 6e^- \rightarrow 2Fe$

Add and cancel $6e^-$ from both sides:

$3Ca + 2Fe^{3+} \rightarrow 3Ca^{2+} + 2Fe$ ✓

Section Review 19.2, Problem 7: Li and Au

Goal: Li is oxidized, Au^{3+} is reduced to Au.

Oxidation: $Li \rightarrow Li^+ + e^-$

Reduction: $Au^{3+} + 3e^- \rightarrow Au$

LCM = 3. Multiply oxidation by 3:

$3Li \rightarrow 3Li^+ + 3e^-$

$Au^{3+} + 3e^- \rightarrow Au$

FINAL: $3Li + Au^{3+} \rightarrow 3Li^+ + Au$ ✓

> Dead battery test: For a voltaic cell to work as a battery, you MUST have one oxidation half-reaction AND one reduction half-reaction. If both half-reactions show oxidation (or both show reduction), no complete circuit is formed and no current flows. Example: $Zn \rightarrow Zn^{2+} + 2e^-$ and $Cd \rightarrow Cd^{2+} + 2e^-$ — both oxidation, so this does NOT form a functional battery.

11 Salt Bridge — Purpose & Mechanism

What is a salt bridge?

A salt bridge is a tube filled with electrolytic gel (or a porous membrane) that physically connects the two half-cells of a voltaic cell without allowing the bulk solutions to mix.

Why is a salt bridge needed?

As the two half-reactions proceed, the charge balance in each half-cell is disrupted:

• In the anode half-cell: cations (like Zn^{2+}) are being produced. The solution becomes positively charged, which repels further oxidation.

• In the cathode half-cell: cations (like Cu^{2+}) are being consumed. The solution becomes negatively charged, which disrupts further reduction.

Without a salt bridge, these charge imbalances quickly stop the reaction. The salt bridge allows ions to flow between the two solutions to maintain electrical neutrality in each half-cell, keeping the reaction going and the current flowing.

What the salt bridge does NOT do:

It does NOT allow the bulk solutions to mix. It only allows small ions to migrate through the gel. This is important because mixing the two solutions could cause direct chemical reaction between them, bypassing the electrode circuit.

From the lab (Mini Lab: Observing a Voltaic Cell):

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Setup: Zn electrode in ZnSO₄ solution | Cu electrode in CuSO₄ solution

Without salt bridge: voltmeter reads 0 (no circuit, no ion flow)

With salt bridge: voltmeter reads ≈1.1 V (circuit complete)


Half-reactions in the lab:

Anode (Zn): Zn → Zn²⁺ + 2e⁻

Cathode (Cu): Cu²⁺ + 2e⁻ → Cu

Overall: Zn + Cu²⁺ → Zn²⁺ + Cu
```

12 Batteries & Fuel Cells

Type	Name	How it works / Key details
Rechargeable	Lead-acid storage battery	Lead electrodes in sulfuric acid solution. Used in cars. Can be recharged by reversing the direction of current.
Rechargeable	Lithium-ion battery	Carbon anode and metal oxide cathode. Li^+ ions travel through a liquid electrolyte. Powers smartphones, laptops, and electric vehicles. Research goal: replace liquid electrolyte with solid-state electrolyte for safety (liquid is flammable). Challenge: solids cannot flow, so ensuring full contact around the electrodes is difficult.
Non-rechargeable	Zinc-carbon dry cell	Classic household battery. Zinc anode, carbon cathode, ammonium chloride paste electrolyte. Inexpensive but limited lifespan.
Non-rechargeable	Alkaline battery	Similar to zinc-carbon but uses an alkaline (basic) electrolyte (potassium hydroxide). Longer shelf life and higher energy density than zinc-carbon.
Special	Fuel cell	Generates electricity from a continuous supply of fuel (commonly hydrogen) and an oxidant (oxygen). Does not deplete like a regular battery — electricity is produced as long as fuel is supplied. Only byproduct of H_2/O_2 fuel cell is water. Used in spacecraft and some vehicles.

13 Practice Problems — Section 19.1 Review

All problems and answers directly from the Chapter 19 slide deck.

Problem 1 DEFINE

What is a redox reaction?

Answer:

A redox reaction is one in which oxidation and reduction both occur; that is, there is a transfer of electrons between substances in the reaction.

Problem 2 DEFINE

Define oxidation.

Answer:

Oxidation is the process by which a chemical substance loses electrons.

The oxidation number of the substance becomes more positive.

Problem 3 APPLY — ASSIGN OXIDATION NUMBERS

Write the oxidation number for each element or polyatomic ion in each compound.

a. F_2 b. $Cu(NO_3)_2$ c. Cr^-

Answer:

a. F_2: F = 0 (free element)

b. $Cu(NO_3)_2$: Cu = +2, NO_3^- = −1 each

c. Cr^-: Cr = −1 (monatomic ion charge)

Problem 4 APPLY — ASSIGN OXIDATION NUMBERS

Write the oxidation number for each element in each compound.

a. $Fe(OH)_2$ b. H_2S c. CH_2O_2

Answer:

a. $Fe(OH)_2$: Fe = +2, OH^- = −1 each (O = −2, H = +1)

b. H_2S: H = +1, S = −2

c. CH_2O_2: C = +2, H = +1, O = −2

Problem 5 TRUE OR FALSE

Oxidation always occurs before reduction. True or False?

Answer:

FALSE.

Oxidation and reduction always occur SIMULTANEOUSLY. You cannot have one without the other.

The electrons lost by the oxidized substance are immediately gained by the reduced substance.

Problem 6 IDENTIFY — REDOX OR NOT?

For each equation, determine whether the reaction is a redox reaction. For each redox reaction, identify which element is oxidized and which is reduced.

a. $2KBr(aq) + Cl_2(g) \rightarrow 2KCl(aq) + Br_2(l)$ b. $2NaOH(aq) + CuCl_2(aq) \rightarrow Cu(OH)_2(s) + 2NaCl(aq)$ c. $Mg(s) + I_2(g) \rightarrow MgI_2(s)$ d. $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$

Answer:

a. YES — redox. Bromine is oxidized (Br: −1 → 0), chlorine is reduced (Cl: 0 → −1).

b. NO — not a redox reaction. No oxidation numbers change (double-displacement).

c. YES — redox. Magnesium is oxidized (Mg: 0 → +2), iodine is reduced (I: 0 → −1).

d. YES — redox. Carbon is oxidized (C: −4 → +4), oxygen is reduced (O: 0 → −2).

Problem 7 ANALYZE — ALL PARTS

In the redox equation $2K + Br_2 \rightarrow 2KBr$, answer all parts:

Answer:

a. Which element is reduced? → Bromine (Br: 0 → −1)

b. Which element is oxidized? → Potassium (K: 0 → +1)

c. What is the reducing agent? → Potassium (it is oxidized)

d. What is the oxidizing agent? → Bromine (it is reduced)

e. Which element loses electrons? → Potassium

f. Which element gains electrons? → Bromine

g. How many electrons per K–Br pair? → 1 electron. Total for reaction? → 2 electrons total.

Problem 8 | BALANCE — OXIDATION-NUMBER METHOD

Balance each equation using the oxidation-number method.

a. $SnCl_4 + Fe \rightarrow SnCl_2 + FeCl_3$ b. $SO_2 + Br_2 + H_2O \rightarrow HBr + H_2SO_4$ c. $CO + I_2O_5 \rightarrow I_2 + CO_2$

Answer:

a. $3SnCl_4 + 2Fe \rightarrow 3SnCl_2 + 2FeCl_3$

Sn: +4 → +2 (reduced, −2), Fe: 0 → +3 (oxidized, +3). LCM=6: ×3 Sn, ×2 Fe.

b. $SO_2 + Br_2 + H_2O \rightarrow 2HBr + H_2SO_4$

S: +4 → +6 (oxidized, +2), Br: 0 → −1 (reduced, −1). ×1 S, ×2 Br.

c. $5CO + I_2O_5 \rightarrow I_2 + 5CO_2$

C: +2 → +4 (oxidized, +2), I: +5 → 0 (reduced, −5). LCM=10: ×5 C, ×2 I (gives 5CO).

Problem 9 | EXPLAIN

Is either of these a redox reaction? a. $2KClO_3 \rightarrow 2KCl + 3O_2$ b. $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

Answer:

a. YES — this IS a redox reaction.

Cl: +5 (in $KClO_3$) → −1 (in KCl) — REDUCED.

O: −2 (in $KClO_3$) → 0 (in O_2) — OXIDIZED.

b. NO — NOT a redox reaction. Na: +1→+1, Cl: −1→−1, Ag: +1→+1, N: +5→+5, O: −2→−2.

This is a precipitation reaction (double-displacement).

14 Practice Problems — Section 19.2 Review

Problem 10 | DEFINE

What is electrochemistry?

Answer:

Electrochemistry is the field of chemistry that studies two related processes: (1) a spontaneous redox reaction that is used to produce electricity, and (2) electricity that is used to cause a nonspontaneous redox reaction to occur.

Problem 11 COMPARE

Name the two types of electrochemical cells and explain the difference between them.

Answer:

The two types are VOLTAIC cells and ELECTROLYTIC cells.

Voltaic cell: the redox reaction is spontaneous and produces electricity (chemical energy → electrical energy). Example: a battery.

Electrolytic cell: the redox reaction is nonspontaneous and requires electricity to occur (electrical energy → chemical energy). Example: electroplating.

Problem 12 EXPLAIN

Why is a special process (electrolysis) necessary to purify active metals such as aluminum and sodium, when other metals can be more easily purified from their ores?

Answer:

The atoms of active metals are held in stronger chemical bonds than other metals.

Also, there are fewer elements that are active enough to replace these elements in the compounds in which they are found.

This means standard chemical reduction cannot work for these metals — electrolysis (supplying electrical energy) is required to break those strong bonds.

Problem 13 ANALYZE — VOLTAIC CELL ELECTRODE

In a VOLTAIC cell: what is the electric charge on the anode? What type of ion migrates toward the anode? What process (oxidation or reduction) occurs at the anode?

Answer:

Charge on anode in voltaic cell: NEGATIVE (−).

Ion type migrating toward the anode: ANIONS (negatively charged ions).

Process at the anode: OXIDATION.

Problem 14 EXPLAIN

Why is a salt bridge used in voltaic cells?

Answer:

As the two half-reactions occur, ions accumulate in the solutions in each half-cell.

In the anode half-cell, cations (like Zn^{2+}) build up, making it positive.

In the cathode half-cell, cations (like Cu^{2+}) are consumed, making it negative.

This charge accumulation would stop the reaction from occurring.

A salt bridge allows the flow of ions between the two electrolytic solutions to maintain electrical neutrality, but prevents the solutions from mixing.

This keeps the reaction going and maintains the flow of electric current.

Problem 15 EVALUATE

Would the following two electrode half-reactions make a functional battery? Explain.

$Zn \rightarrow Zn^{2+} + 2e^-$ and $Cd \rightarrow Cd^{2+} + 2e^-$

Answer:

NO — these would NOT make a functional battery.

Both reactions are OXIDATION half-reactions.

For a working battery (voltaic cell), you must have one oxidation half-reaction AND one reduction half-reaction to form a complete redox reaction.

Because both Zn and Cd are being oxidized, there is no cathode, no reduction, and therefore no complete circuit — no current would flow.

Problem 16 WRITE HALF-REACTIONS

Write the half-reactions and the overall redox reaction for lithium being oxidized and gold(III) ions being reduced to gold atoms.

Answer:

Oxidation half-reaction: $Li \rightarrow Li^+ + e^-$

Reduction half-reaction: $Au^{3+} + 3e^- \rightarrow Au$

Balance electrons (LCM = 3): multiply oxidation ×3:

$3Li \rightarrow 3Li^+ + 3e^-$

$Au^{3+} + 3e^- \rightarrow Au$

Cancel electrons and combine:

FINAL: $3Li + Au^{3+} \rightarrow 3Li^+ + Au$ ✓

Problem 17 APPLY — LITHIUM-ION BATTERY

Summarize how a lithium-ion battery works.

Answer:

A lithium-ion battery has a carbon electrode and a metal oxide electrode, where the half-reactions of a redox reaction take place.

Lithium ions (Li^+) travel through a liquid electrolyte during the reaction.

This movement of ions creates an electric current that is used to power devices.

The battery is rechargeable: reversing the current direction reverses the reaction.

Problem 18 ANALYZE – SOLID STATE ELECTROLYTE

What is the challenge of using a solid-state electrolyte in a lithium-ion battery?

Answer:

All chemical reactions require that the reactants come in contact with each other.

Because solids cannot flow (unlike liquids), it is more difficult to ensure that the solid electrolyte makes full contact all around the electrode surfaces.

Poor contact reduces efficiency and makes the battery less effective.

Researchers are working to engineer solid electrolytes with sufficient contact.

All content sourced directly from Chemistry 5th Ed. Chapter 19 slide deck. Symbols rendered with DejaVu Unicode font: → (yields), ⇌ (equilibrium), · (separator).